

Smart Meter Dataset Architecture & Forensic Analysis

Evaluation, physical provenance, and data verification gates for dual-system load forecasting & clustering.

1. Candidate Dataset Evaluation & Suitability

SELECTION MATRIX

To support both **System A (Demand Forecasting via Regression)** and **System B (Customer Segmentation via K-Means)**, candidate corpora must exhibit persistent meter keys (`meter_id`), continuous high-frequency temporal stamps, and multi-gigabyte volume justifying distributed Apache Spark execution.

Dataset	Scale & Scope	Resolution	System A (Forecasting)	System B (Clustering)	Spark Scale Fit
CEEW Mathura & Bareilly	~100 meters, 20M rows	3-minute (2019–21)	Optimal Electrical parameters	Optimal Distinct consumer vectors	Strong: Uncompressed CSVs
Smart Meters in London	5,567 meters, 167M rows	30-minute (2011–14)	Optimal Weather-linked	Optimal Rich socio-demographics	Very Strong: ~11 GB footprint
Building Data Genome 2	3,053 meters, 1,636 sites	1-hour (2016–17)	Good Non-residential	Moderate Commercial archetypes	Moderate: ~53M records
UCI Household Electric	1 meter, 2.07M rows	1-minute (2006–10)	Good Single home series	Unusable Single observation unit	Weak: Runs in pandas RAM

2. End-to-End System Dual-Architecture

PIPELINE VISION

REAL-WORLD ELECTRICITY PHENOMENA (Smart Meter AMI Telemetry) | ▼ CEEW MATHURA & BAREILLY (20M Row 3-m

System A: Demand Forecasting Target: Numerical energy consumption prediction (Y_{t+h}). Inputs: Dynamic temporal lags ($t-1, t-2, t-24$), cyclical time vectors (sine/cosine of hour/day), and power quality conditions (V, I, f). Model Suite: Spark Linear Regression, Decision Trees, and GBT Regressors.	System B: Customer Segmentation Target: Discrete behavioral customer segment assignments. Inputs: Longitudinal behavioral aggregations (daily mean kWh, Peak-to-Average Ratio, day/night usage ratio, outage exposure). Model Suite: MLLib K-Means with Silhouette & Elbow optimization.
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Collected by the **Council on Energy, Environment and Water (CEEW)** in collaboration with academic researchers, this corpus records real-world supply reliability, brownout frequency, and consumption dynamics under MVVNL (Bareilly) and DVVNL (Mathura) utilities in Uttar Pradesh, India. Data is public under the **CC0 1.0 Universal License** (Harvard Dataverse DOI: 10.7910/DVN/GOCHJH).

Archive File Name	Format	Temporal Coverage	Operational Engineering Role
CEEW - ReadMe.txt	Text	Metadata	Variable dictionary, equipment notes, and institutional provenance.
CEEW - Smart meter data Mathura [2019-2021]	CSV	May 2019 – Oct 2021	Primary raw 3-minute interval telemetric readings for Mathura district.
CEEW - Smart meter data Bareilly [2019-2021]	CSV/TAB	May 2019 – Oct 2021	Primary raw 3-minute interval telemetric readings for Bareilly district.
CEEW - Smart meter data [District] Aggregated	TAB	2019 – 2021 Daily	Reference totals for validating distributed PySpark SQL aggregation logic.

4. Physical Semantics, Units & Electrical Formulations

PHYSICS LAYER

A single raw row forms an atomic physical observation: Row = $\langle \text{meter}, \text{Timestamp}, \Delta E, V_{\text{RMS}}, I_{\text{RMS}}, f \rangle$.

Interval Active Energy (t_{kWh})

Active energy drawn strictly during the 3-minute interval ($\Delta t = 3/60 = 0.05 \text{ h}$). It is **incremental, non-cumulative**:

$$E = P \times 0.05 \text{ kWh} \Leftrightarrow 0.050 \text{ kWh} \equiv 1.0 \text{ kW load}$$

Grid Voltage & Frequency

RMS Voltage (nominal 230V $\pm 6\%$) and Grid Frequency (nominal 50.0 Hz). Real-world supply exhibits frequent brownouts ($< 180\text{V}$) and severe dips.

$$\text{Active Power: } P = V_{\text{RMS}} \times I_{\text{RMS}} \times \cos(\varphi)$$

5. Physics of Zeros: Outages vs. Standby

INTEGRITY RULE

Zero-load observations represent two fundamentally distinct physical realities that must not be conflated during preprocessing:

CRITICAL DATA ENGINEERING DISTINCTION:

- **Voluntary Inactivity:** $V \approx 230\text{V}, f \approx 50\text{Hz}, I = 0\text{A}, t_{\text{kWh}} = 0 \rightarrow$ Grid fully energized; residents asleep or house vacant. Retain numerical 0.0.
- **Grid Outage (Blackout):** $V = 0\text{V}, f = 0\text{Hz}, I = 0\text{A}, t_{\text{kWh}} = 0 \rightarrow$ Feeder trip, load shedding, or line fault. Must be explicitly flagged as $\text{is_outage} = 1$ to prevent models from learning false zero-demand elasticity.

TEMPORAL CONTINUITY AUDIT: Theoretical (100 meters x 914 days x 480 intervals/day) $\approx 43,872,000$ Expect

6. Downsampling & Feature Engineering Formulation

TRANSFORMATIONS

Because raw 3-minute readings are dominated by high-frequency appliance switching noise (refrigerators, water pumps), data must be regularized into 1-hour intervals for reliable forecasting and customer vectorization.

Hourly Resampling (System A)

Aggregate interval measurements over clock-hour buckets:

$$Y_{m,T} = \sum t_kWh \mid V_{m,T} = \text{mean}(V) \mid I_{m,T} = \text{mean}(I)$$

- **Auto-Regressive Lags:** $Y_{T-1}, Y_{T-2}, Y_{T-24}, Y_{T-168}$
- **Rolling Stats:** 24-hour mean (μ_{24}) and standard deviation (σ_{24})
- **Calendar Cycles:** sin/cos encodings for hour-of-day and day-of-week.

Behavioral Vectors (System B)

Collapse longitudinal series into a d -dimensional vector:

$$PAR_m = \max(Y_{m,T}) / \text{mean}(Y_{m,T})$$

- **Volume:** Average daily energy consumption (E_m)
- **Volatility:** Coefficient of variation ($CV = \sigma / \mu$)
- **Diurnal Ratio:** Daytime (06:00–18:00) vs. Night energy
- **Stress Index:** Total outage hours and brownout fraction ($<200V$).

7. Production Pipeline Architecture & Data Contract

PYSPARK IMPLEMENTATION

[Raw CSV Directory] (Immutable Source Archive) | ▼ (Explicit StructType Schema Enforcement – inferSchema)

8. Verification Gates Checklist

PRE-FLIGHT REVIEW

- **Gate 1 (Provenance Verification):** Verified Harvard Dataverse source under CC0 Public Domain Dedication.
- **Gate 2 (File Inventory & Checksum):** Validated 8 primary district CSVs/TABs with SHA-256 integrity logs.
- **Gate 3 (Schema Contract):** Instantiated explicit StructType without expensive dual-pass type inference.
- **Gate 4 (Data Quality Audit):** Checked for negative energy, impossible voltages ($>300V$), and duplicate keys.
- **Gate 5 (Temporal Regularization):** Mapped missing observation intervals; isolated true zero-load from grid blackouts.
- **Gate 6 (Modeling Suitability):** Ensured time-series data leakage is prevented via sequential temporal train/test split.
- **Gate 7 (Dataset Freeze):** Serialized final validated parquet datasets for MLlib pipeline consumption.